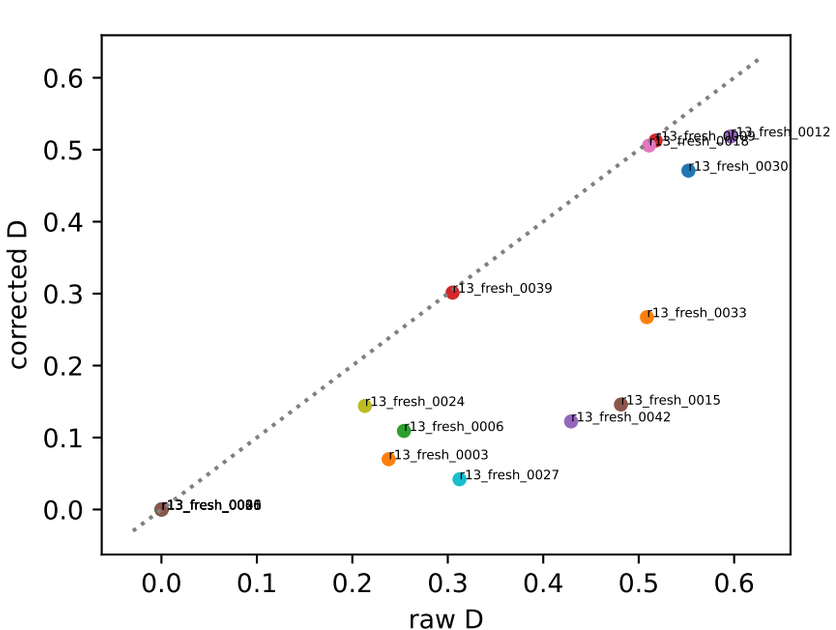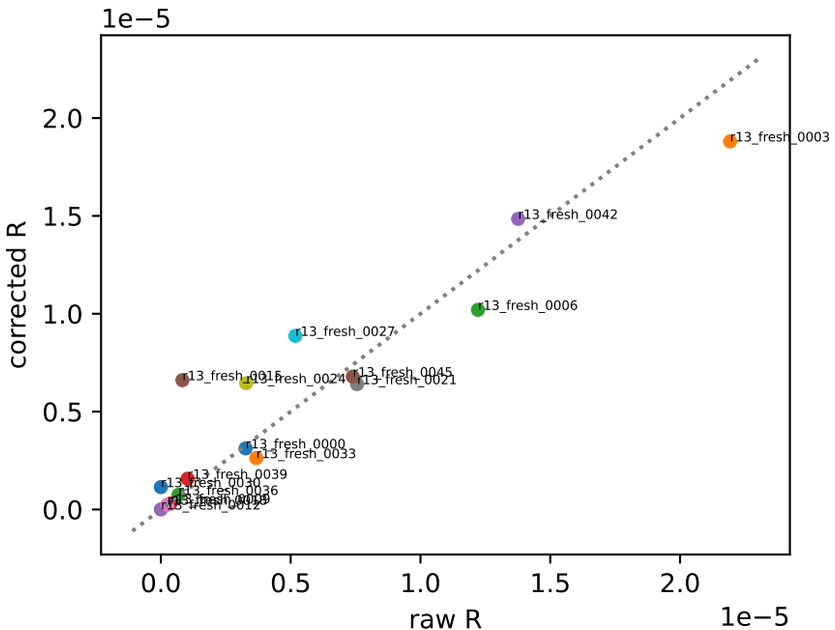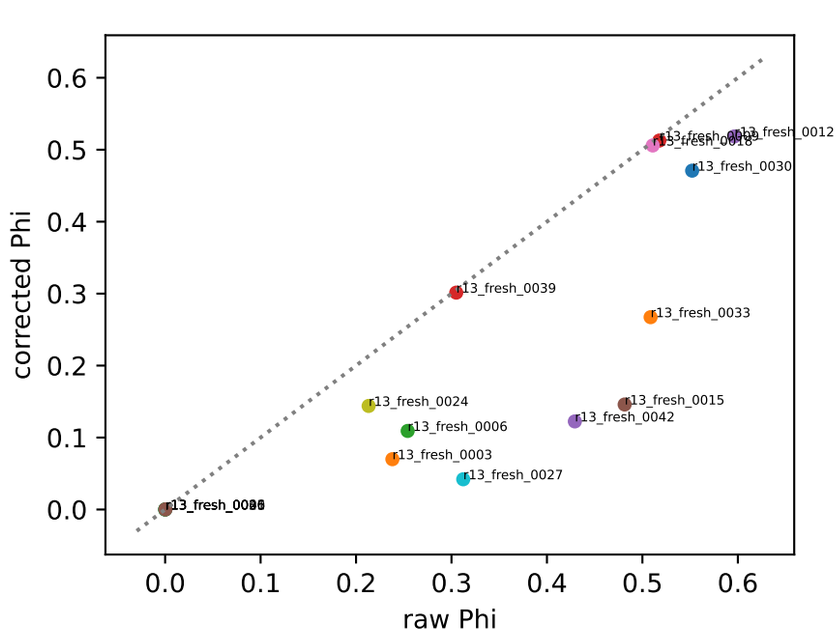ap corrector decomposition; lower objective need not improve material